

Hidden Permutation

Time limit: 1.00 s

Memory limit: 512 MB

There is a hidden permutation $a_1, a_2, \dots, a_n$ of integers $1, 2, \dots, n$. Your task is to find this permutation. To do this, you can ask questions: you can choose two indices i and j and you will be told if $a_i < a_j$.

Interaction

This is an interactive problem. Your code will interact with the grader using standard input and output. You should start by reading a single integer n : the length of the permutation.

On your turn, you can print one of the following:

- " $? \ i \ j$ ", where $1 \leq i, j \leq n$: ask if $a_i < a_j$. The grader will return YES if $a_i < a_j$ and NO otherwise.
- " $! \ a_1 \ a_2 \ \dots \ a_n$ ": report that the hidden permutation is $a_1, a_2, \dots, a_n$. Your program must terminate after this.

Each line should be followed by a line break. You must make sure the output gets flushed after printing each line.

Constraints

- $1 \leq n \leq 1000$
- you can ask at most 10^4 questions of type $?$

Example

```
3
? 3 2
NO
? 3 1
YES
! 3 1 2
```

Explanation: The hidden permutation is $[3, 1, 2]$. The first question asks if $a_3 < a_2$ which is false, so the answer is NO. The second question asks if $a_3 < a_1$ which is true, so the answer is YES.